completed 2 adjuvant cycles after RT without dose modification. Except for 3 N0-patients, 26 (70.3%) patients reached partial response to induction chemotherapy, and 11 (29.7%) had complete response, as far as regional nodes were concerned. For a median follow-up of 31 months, 2 patients developed recurrences in the nasopharynx, 1 had both hepatic and multiple bone metastases, and 1 had hepatic metastasis. The former-mentioned 3 patients were dead, the latter one had transcatheter arterial embolization and is alive at the time of analysis. 3-year locoregional progression-free, distant metastases-free, disease-free and overall survival rates were 93.8%, 94.5%, 88.5% and 90.0%, respectively.

Conclusions: This GP regimen demonstrated an impressive anti-tumoral activity in locally advanced NPC, with a favorable toxicity profile. The current approach may improve the outcome of patients with locally advanced NPC and merits further investigation.

Author Disclosure: D. Ou, None; X. He, None; C. Hu, None; G. Zhu, None; H. Ying, None; Y. Wu, None.

2498 Choroidal Melanoma Treated with Ruthenium-106 Plaque Brachytherapy: Effectiveness and Complications

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Purpose/Objective(s): To evaluate the clinical outcome of small and medium-sized choroidal melanomas treated with Ruthenium-106 plaque brachytherapy.

Materials/Methods: A retrospective study was performed in 94 patients with choroidal melanoma treated with Ruthenium-106 plaque brachytherapy at a single institution between January 2004 and September 2008. The minimum follow-up was 6 months and median age was 60 years (18–84 years). The dose prescribed was 85 – 100 Gy to depth of 5 mm or to the tumor apex, when the height was greater than 5 mm. The tumors had a median height, including sclera, of 4.33 mm (2.25 – 6.95 mm) and a median largest basal dimension of 9.25 mm (3.76 – 15.79 mm). The clinical follow-up included regular ocular examination (fundoscopy, ultrasonography, analysis of visual acuity) and systemic assessment.

Results: Eighty three patients were evaluated (11 patients were not eligible) with a median follow-up time of 22 months (6 – 59 months). The median apical and scleral doses were, respectively, 100.8 Gy (43 – 207 Gy) and 307.4 Gy (251 – 420 Gy), with a median treatment time of 72 hours (48 – 288 hours). There were 6 local failures and 1 distant failure. During the follow-up period, there was one death due to other confirmed malignance (renal cancer). There were 6 enucleations: 4 secondary to local failure and 2 secondary to treatment complications (refractory ocular pain and vitreous bleeding). One patient received transpupillary thermotherapy (TTT) 3 months before the brachytherapy. Four patients developed cataracts in the treated eye. Vision stayed the same or improved in 48% of patients, and 61% of patients maintained visual acuity better than 20/200. In our sample, including 81 patients (2 enucleated patients due to treatment complications were excluded), the actuarial local control and progression free survival at 2 and 5 years were, respectively, 98.3%, 91.3% and 90%, 88.7%.

Conclusions: These early results show that Ruthenium-106 plaque brachytherapy is a reliable therapy and offers a high local tumor control rate for small and medium-sized choroidal melanoma. In addition, the majority of patients had favorable visual outcome and low incidence of complications.

Author Disclosure: G.O. Gil, None; D.G. Marconi, None; D.G. Castro, None; L.M. Rebouças, None; H. Balloni, None; A.C.A. Pellizon, None; A. Teston, None; M. Motono, None; M.M.M. Chojniak, None; J.V. Salvajoli, None.

2499 Tumor Volume Predicts Tumor Control and Survival among Patients with Locally Advanced Head and Neck Cancer undergoing Definitive Chemoradiation Therapy

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Purpose/Objective(s): Tumor volume has been established as a significant predictor of outcomes among patients with head and neck cancers receiving radiation therapy alone. This study attempts to add to the existing literature on tumor volume as a prognostic factor among patients receiving chemoradiation therapy.

Materials/Methods: 80 patients who underwent definitive chemoradiation for Stage III–IV squamous cell cancer of the hypopharynx, oropharynx and larynx were identified. All patients had a minimum 18-month follow-up. Data were collected on T-Stage, N-Stage, gross tumor volume, and involved node volume based on treatment planning CT scans; these were correlated to survival and tumor control data obtained from retrospective analysis.

Results: Bivariate regression found primary tumor volume to be a significant predictor for length of survival ($p = 0.022$) and time to recurrence ($p = 0.007$), with smaller volumes leading to better outcomes. A similar analysis for involved lymph node volume did not show a significant correlation between nodal volume and risk of tumor recurrence or death. Tumor volume was a continuous predictor of tumor control and survival, with 35cc appearing to be the point at which there was the most significant difference in outcomes. Patients with volumes <35cc had a significantly better prognosis, in terms of 5-year overall survival (84% vs. 43%,$p = < 0.001$), progression-free survival (61% vs. 33%, $p = 0.004$), and time to progression (71% vs. 41%, $p = 0.004$). The tumor volume data was also analyzed against age, gender, T-Stage, N-Stage, and site of disease in a proportional hazards model to determine factors with predictive value for risk of failure. Tumor volume >35cc was the best predictor of recurrence (HR = 4.7, CI = 1.9, 11.6; $p = 0.001$), followed by site (HR = 2.1, CI = 1.3, 3.5; $p = 0.004$). Similarly, tumor volume (HR = 10.0, CI = 2.9, 35.1; $p = <0.001$) and site (HR = 2.3, CI = 1.2, 4.4; $p = 0.014$) were also strong predictors for overall survival. The other factors did not show a statistically significant effect on the risk of failure. Additionally, gross tumor volume was a predictor for the site of failure. Analysis of variance found tumors that failed locoregionally to be on average 21.6cc larger than tumors that did not fail ($p = 0.028$).